

LIFE CYCLES OF STARS

Reference

See pp. 3 - 7 of Science Aspects Book 4.

Background

There are three possible pathways for stars to follow in their lives. All stars come from a similar start point, but it is their size that ultimately determines their final destiny. Some become black dwarfs, some neutron stars, and some black holes.

Task

Your task is to use the Internet to find useful information that can provide *simple answers* to the following questions. Several useful sources are listed below.

- Good information - probably the *best site* to get information.

<http://www.enchantedlearning.com/subjects/astronomy/stars/lifecycle/giant.shtml>

- Activity on aging photos of stars.

<http://btc.montana.edu/ceres/html/LifeCycle/stars1.html> - activity2

- Photos from Hubble.

<http://hubblesite.org/newscenter/newsdesk/archive/releases/1999/20/>

- Power point - black holes.

<http://imagine.gsfc.nasa.gov/docs/teachers/blackholes/blackholes.html>

- Power point -life cycles of stars.

<http://imagine.gsfc.nasa.gov/docs/teachers/lifecycles/stars.html>

You may find others by using Metacrawler (<http://www.metacrawler.com/>) and doing a global search on "life cycles of stars".

Questions

1. (a) What is a *nebula*?

• Cloud of gas (H) + dust.

- (b) What happens in a nebula?

• Gravity pulls gas together.
• Friction heats it up.

2. How do *protostars* form?

- Gravity compresses the gases + heats it.
- Fusion ($H \rightarrow \text{Helium}$) starts.

3. There are 7 *main types of stars*. What are they and how are they classified?

Types:

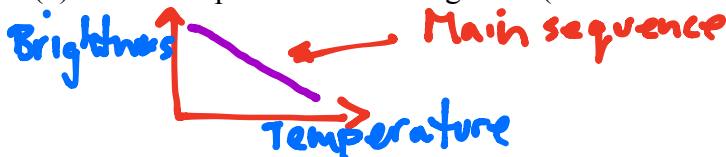
O B A F G K M
blue white yellow red.

Classification:

4. (a) What does the Hertzsprung-Russell diagram show about stars?

- Life cycle of stars.

(b) What is plotted on this diagram? (Hint: What labels appear on the axes?)



5. (a) What is a "main sequence star" according to this diagram?

- Stars that are "normal"
- Not dying!

(b) Is our Sun a member of this group?

Yes

6. How are stars grouped according to their size? (There are *three groups* - think of their possible life cycles.)

(i) $< 1.5 \times \text{Sun}$

(ii) $1.5 - 3.0 \times \text{Sun}$

(iii) $> 3.0 \times \text{Sun}$

7. Which stars become:

(a) black dwarfs? *About our Sun size.*

(b) neutron stars? $1.5 \rightarrow 3.0 \times$ Our sun

(c) black holes? $> 3.0 \times \text{Our Sun}$

8. What are the characteristics of the three stars listed in Q 7?

Black dwarf: **dead, no light or heat**

Neutron star: **very dense, strong magnetic field**

Black hole: **very small, dense, extreme gravity**

9. (a) What will our star become when it is near the end of its life?

white dwarf $\rightarrow$ black dwarf

(b) Describe what will happen to our Sun in the next 5 billion years - use dot points.

Sun $\rightarrow$ red giant $\rightarrow$ nova $\rightarrow$ white dwarf
 $\swarrow$
 black dwarf

10. (a) What is the difference between a *nova* and a *supernova*?

Nova - Small explosion.
Supernova - Massive explosion.

- (b) Which stars will supernova?

$> 1.5 \times \text{Sun}$

11. Find a *one-line description* for the following types of stars.

- (a) Yellow dwarf: small, yellow star
(b) Red dwarf: small, cool star
(c) White dwarf: small, hot star
(d) Brown dwarf: small, no nuclear fusion
(e) Red giant: old, cool, large
(f) Blue giant: huge, hot, blue colour
(g) Supergiant: largest star
(h) Pulsar: spinning neutron star

12. (a) What is a quasar?

Star-sized energy source - gives off lots of radiation.

- (b) Is it a star?

No - it's a feeding black hole